2018B F=ma Exam: Problem 7

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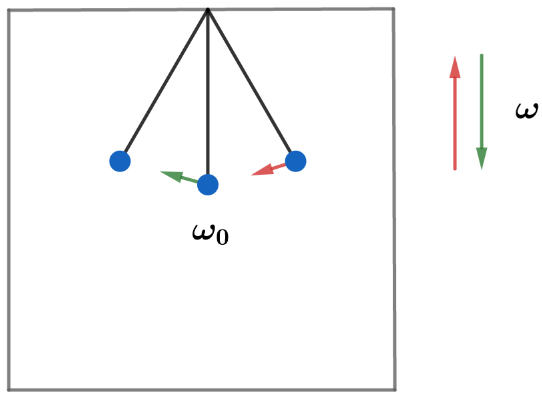

To maximize the final amplitude, force should be applied in the same direction as the pendulum's motion. Thus, the box should accelerate upwards when the mass is traveling down and accelerate downwards when the mass is traveling up.

This is clear in the frame of the box by following the direction of the fictitious force. Hence, the box completes one oscillation for every half oscillation of the pendulum:

$$\omega = 2\omega_0 = 2\sqrt{\frac{g}{L}} = \sqrt{\frac{4g}{L}}$$

so the answer is $\boxed{\text{A}}$.